

Miah Lynne Manning

Biology Department | MIT-WHOI Joint Program | Woods Hole, MA 02543 | miahm@mit.edu

EDUCATION

Ph.D. in Biological Oceanography, **MIT-WHOI Joint Program** June 2023 - present
Advisor: Dr. Harriet Alexander
B.A. in Biological Sciences, **Rutgers University** – New Brunswick 2020 - 2023
School of Environmental and Biological Sciences Honors Program, Minor in Marine Science, Cumulative GPA:
4.0, *Summa cum laude*

RESEARCH EXPERIENCE

Microbial Diversity Advanced Research Training Course July - Aug. 2024
Marine Biological Laboratory (MBL), Directors: Dr. Scott Dawson and Dr. John Spear
Rutgers Center for Ocean Observing Leadership (RUCOOL), Undergraduate researcher 2021 - 2023
Advisor: Dr. Oscar Schofield, Undergraduate Honors Thesis: *Characterizing the variability of phytoplankton on the West Antarctic Peninsula*
Waksman Institute of Microbiology, Laboratory Assistant 2022 - 2023
Mote Marine Laboratory & Aquarium, NSF REU Internship June - Aug. 2022
Phytoplankton Ecology Program, Advisor: Dr. Vince Lovko, Project title: *Characterizing the photophysiology of Karenia brevis following exposure to bloom mitigation compounds*

TEACHING EXPERIENCE

Teaching Trainee, MIT-WHOI Joint Program, Woods Hole, MA Feb. – May. 2026
7.470 Biological Oceanography; Conducted recitation sections, graded problem sets and daily assignments, advised professors on student performance, and taught a guest lecture titled “Chemical Signaling & Microbial Interactions”.
Co-Teaching Trainee, MIT-WHOI Joint Program, Woods Hole, MA Sept. – Dec. 2025
12.739 Marine Microbiology and Biogeochemistry; Led weekly recitations and taught a guest lecture titled, “The Phycosphere Under Stress: Investigating algal-bacteria dynamics in a changing ocean”.
Joint Program Summer Math Review Instructor & Program Leader, 2025, 2026
MIT-WHOI Joint Program, Woods Hole, MA; Probability & Statistics; Taught a 1.5 hr. lecture, provide notes and code examples; Program lead: organize and train lecturers, assign topics, reserve rooms, and advertise courses to students
Guest lecturer, Cape Cod Community College, Barnstable, MA 23 April 2024
BIO109 Survey of Biology, Title: “Marine Microbial Diversity”
Graduate Student Teaching Certificate Program, MIT, Cambridge, MA 2024 – 2026
MIT Teaching and Learning Laboratory; Training: constructed a class syllabus, problem sets and exam questions, taught active-learning exercises, and gave a lecture with personalized feedback

PUBLICATIONS

In review

Fisher, B. J., Seidel, M., Waite, N., Manning, M., Cappola, M., Dittmar, T., Schofield, O., Poulton, A. J., and Henley, S. F. Southern Ocean phytoplankton community structure is associated with variability in the molecular composition of dissolved organic matter. *submitted* to Journal of Geophysical Research: Oceans.

In preparation

Manning, M., Shapiro, S., and Alexander, H. Genetic signatures of phycosphere specialization in the marine bacteria genus *Marinobacter*. *In preparation* for mSystems. Expected submission date: May 2027.

Manning, M., Krinos, A., Shapiro, S., Kopczynski, S., Hamilton, R., Johnson, W., Mars Brisbin, M., and Alexander, H. Strain-specific osmolyte and phycosphere responses of *Gephyrocapsa huxleyi* to nitrogen limitation and warming. *In preparation* for ISME communications. Expected submission date: January 2027.

SELECTED PRESENTATIONS

Manning, M., Shapiro, S., Chan, B., Alexander, H. Signatures of phycosphere interactions hidden within the *Marinobacter* pangenome. Marine Microbes Gordon Research Conference (GRC) and Gordon Research Seminar (GRS), Barcelona, Spain, June 2026.

Manning, M., Krinos, A., Shapiro, S., Hamilton, R., Mars Brisbin, M., Johnson, W., and Alexander, H. Strain-specific

effects on *Gephyrocapsa huxleyi* bacterial communities under temperature and nitrogen co-stress. Association for the Sciences of Limnology and Oceanography (ASLO) Aquatic Sciences Meeting, Charlotte, NC, March 2025.

Manning, M., V. Lovko, S. Chakraborty. Characterizing the photophysiology of *K. brevis* following exposure to bloom mitigation compounds. Association for the Sciences of Limnology and Oceanography (ASLO) Aquatic Sciences Meeting, Palma de Mallorca, Spain, June 2023.

Manning, M., V. Lovko, S. Chakraborty. Characterizing the photophysiology of *K. brevis* following exposure to bloom mitigation compounds. 11th National Harmful Algal Bloom (HAB) Symposium, Albany, NY, October 2022.

Honorable Mention "Best Poster Presentation by a Student"

RESEARCH CRUISES

Abundant Osmolytes; HRS 26-06, R/V Hugh R. Sharp	8-24 July 2026
Chief Scientist: Dr. Erin McParland	
Joint Program Student Cruise; AR96, R/V Neil Armstrong	24-26 Oct. 2025
Served as biology sampling leader; Co-Chief Scientists: Lilli Enders and Katherine Squires	
Joint Program Student Cruise; AR80, R/V Neil Armstrong	17-19 Nov. 2023
Co-Chief Scientists: Kate Lane, Lina Taenzer, Lukas Taenzer	
Palmer Long Term Ecological Research Cruise (Pal-LTER), LMG 23-01	Dec. 2022 – Feb. 2023
R/V Laurance M. Gould, Chief Scientist: Dr. Carlos Moffat	

AWARDS

Research Fellowships

Grassle Fellowship Fund, MIT-WHOI Joint Program \$7,400, Aug. 2025

Distinctions and Honors

Matthew Leydt Society, Rutgers University, inductee May 2023

Phi Beta Kappa, Rutgers University, inductee April 2023

Scholarships

Patricia A. Case Endowed Scholarship, Marine Biological Laboratory July 2024

Cook/CAES Scholarship, Rutgers University Sept. 2022

Alampi Phillip Scholarship, Rutgers University Sept. 2022

George H. Cook Scholars Program, Rutgers University Dec. 2021

Hamo Hachnasarian Scholarship, Rutgers University Sept. 2021

OUTREACH

The Synergy II Project – Scientist	Jan 2026 - present
Collaboration with educator and artist John Wolf to create an art installation describing the roles of phytoplankton in the marine environment.	
Woods Hole Science Stroll – Volunteer	Aug. 2024, 2025, 2026
Zephyr Marine – Volunteer	Sept. 2023 – present
Geoscience Education Mentorship (GEM) - Mentor	Sept. 2023 – present
Letters to a Pre-Scientist – Mentor	Sept. 2023 – May 2024
Lower Raritan Watershed Partnership (LRWP) – Volunteer	Fall 2022 – May 2023

ORGANIZATIONS & COMMITTEES

Ocean Carbon & Biogeochemistry (OCB) Intercomparing metabarcoding of marine microbial communities working group	July 2026 - present
Broader Impacts Group (BIG) – Board Member	Sept. 2024 – present
Joint Program Applicant Support & Knowledgebase – Board Member	Sept. 2023 – present
Joint Program Representative – Department Representative	Oct. 2024- Sept. 2025

SKILLS

Programming: R, shell scripting, Python

Other Computing Skills: High-Performance Computing systems, SLURM scheduler, Adobe Illustrator, Git

Laboratory: Oxford Nanopore library prep and sequencing, RNA/DNA extraction, polymerase chain reaction, gel electrophoresis, imaging flow cytometry, Pulse-Amplitude-Modulation (PAM) fluorometry, fluorescence microscopy, bacterial and algal isolation and culturing